

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Examinations

April - May 2017

Max. Marks: 100

Class: Second Year

Name of the Course: Relational Database Management Systems

Course Code: UCEC404

Duration: 3 Hours

Semester: IV

Branch: Comp

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks
Q 1 (a)	Define the following terms with the help of example. (i) Simple and composite attribute (ii) Entity Set and Weak Entity (iii) Derived Attribute (iv) Single valued and multi valued attribute.	10
Q1 (b)	Describe the overall architecture of DBMS.	10
Q2 (a)	What is transaction? Discuss ACID properties of transaction?	10
	OR	
Q2 (a)	Explain the Two phase Locking protocol.	10
Q2 (b)	Explain following Relational Algebra Operations with example. (i) Natural Join (iii) Generalized Projection (ii) Set Intersection (iv) Division Operation	10
	OR	
Q2(b)	Explain following Relational Algebra Operations with example. (i) Cartesian product (ii) Outer Join (iii) Aggregate Functions (iv) Rename	10
Q3 (a)	What is Normalization? Explain 1NF, 2NF, 3NF and BCNF with Examples.	10
Q3 (b)	Explain Security and Authorizations in DBMS.	10
	OR	
Q3(b)	Explain Data Definition and Manipulation commands in SQL.	10

Q4 (a)	For the following given database, write SQL queries:- Teacher(Tid,Teacher_name,street,city) Teaches(Tid, Sid) Student(Sid, Student_name, city, class) (i) Find Teacher name and city of Teachers who teaches students of class "F.E" and those stays in city "Mumbai" (ii) Find Sid and class of students whose name are not "Rajesh" and "Ritesh" (iii) Find all teacher details whose name starts with 'J'. Assume any values to required attribute.	10
Q4 (b)	What do you mean by deadlock? What are various techniques for deadlock prevention and detection? OR Explain Timestamp ordering protocol.	10
Q4 (b)		10
Q5 (a)	Explain Assertions and Triggers in detail.	10
Q5 (b)	Explain various rules for mapping ER and EER to Relational Model.	10